Lecture 12: The Integral Test

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand when the Integral Test can be used.
- ☐ Use the Integral Test to decide if a series converges or diverges.
- ☐ Use the error bounds to estimate the partial sums.

Nondecreasing Partial Sum

Suppose that $\sum_{n=1}^{\infty} a_n$ is an infinite series with $a_n > 0$ for all n. Then each partial sum is greater or equal to its predecessor.

A series $\sum_{n=1}^{\infty} a_n$ of nonnegative terms converges if and only if its partial sums are bounded from above.

Nondecreasing Partial Sum

Example: The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

The Integral Test

Does the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge?

The Integral Test

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ where N is a positive integer. Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.

The Integral Test

Example: For a real constant p, the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

converges if $p > 1$ and diverges if $p \leq 1$.

The Integral Test

Example: The series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges.

The Integral Test

Example: The series $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges.

Error Estimation

Suppose a series $\sum a_n$ with positive terms converges to S, we want to estimate the size of the remainder R_n by measuring the difference between S and the nth partial sum s_n.

$$R_n = S - s_n = a_{n+1} + a_{n+2} + a_{n+3} + \cdots$$

Bounds for the Remainder in the Integral Test

Suppose $\{a_k\}$ is a sequence of positive terms with $a_k = f(k)$, where f is a continuous, positive, decreasing function of x for all $x \geq n$, and suppose that $\sum a_n$ converges to S. Then the remainder $R_n = S - s_n$ satisfies the inequalities

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_{n}^{\infty} f(x)\,dx.$$

To estimate S, we add the partial sum s_n to each side of the inequalities.

$$s_n + \int_{n+1}^{\infty} f(x)\,dx \leq S \leq s_n + \int_{n}^{\infty} f(x)\,dx.$$

Bounds for the Remainder in the Integral Test

Example: Estimate the sum of the series $\sum \frac{1}{n^2}$ using the previous inequalities and $n = 10$.